RM07

Ranking, Seating Arrangement and Logical Venn Diagrams

The biggest reasoning chapter - 3-5 questions in every RAS Prelims paper. Seating arrangement usually comes as a small set of two or three questions on one common stem, which means one good diagram pays for three marks. Ranking and numeric Venn are single, quick questions and should never be lost.

Three different question families share one chapter because they share one habit: you fix a reference point first, then place everything else against it. Ranking is one formula. Seating is a set of dashes on paper. Venn is two formulas and five standard pictures. None of it needs cleverness, only order.

Ranking, Seating and Venn

Ranking
- Bottom = Total - Top + 1
- Same person: Total = a + b - 1
- Between = Total - left rank - right rank
- Shifting and interchange

Comparison
- One single inequality chain
- Tallest at one end
- Count from the correct end

Linear seating
- Fix an END or the MIDDLE first
- Facing North: their right = your right
- Facing South: everything flips
- Third to the left = two people between

Circular / square
- Facing centre: clockwise = LEFT
- Facing outward: clockwise = RIGHT
- Opposite seat is n/2 away
- Square: diagonal = two corners

Double row
- Row 1 North, row 2 South
- Mirror-image left-right senses
- X's right faces Y's left

Floors / days
- Floor 1 at the bottom
- 'Above' = larger number
- Days as slots 1 to n
- Test both cases of a gap clue

Logical Venn
- All A are B = A inside B
- Some A are B = overlap
- No A is B = separate circles
- Nested vs co-ordinate patterns

Numeric Venn
- Two-set: A + B - both
- Three-set inclusion-exclusion
- Exactly one, exactly two
- Neither = Total - at least one

Three question families, one working habit

This chapter carries three separate things. Ranking asks where a person stands in a row or a merit list. Seating arrangement gives you a row, a circle, a square, a set of floors or a set of days, plus five or six clues, and asks you to place everyone. Logical Venn asks either which picture shows the relation between three classes, or a numeric set problem. What ties them together is the working habit: never start placing until you have found the one clue that fixes a reference point.

- Order and ranking - single-ended and double-ended, shifting, interchange.
- Comparison and ordering puzzles - tallest, heaviest, third from the top.
- Linear seating - single row facing North or facing South.
- Double-row seating - two parallel rows facing each other.
- Circular seating - facing the centre or facing outward; square and rectangular tables.
- Floor puzzles and day or month scheduling puzzles.
- Logical Venn - which diagram represents Doctors, Men and Human beings.
- Numeric Venn - two-set and three-set set problems.

Where marks leak. In ranking, the student forgets the plus one or subtracts it in the wrong place. In seating, the student starts from a weak clue like 'X is not next to Y' and builds three branches instead of one. In circular seating, the student uses left and right as if the persons were facing outward when they are facing the centre. In numeric Venn, the student treats the pairwise figure as if it excluded the people who take all three. Each is a single specific slip, and each is fixable in a minute.

ASKED IN RAS

Seating arrangement has appeared in every RAS Prelims paper, usually as a linked set of two or three questions. Ranking came in 2018 (rank from the bottom in a class of 40 - answer 29th) and 2023 (10th from the front, 15th from the back - answer 24). Circular seating came in 2021 and linear seating in 2024. Logical Venn class-relationship questions came in 2016, 2021, 2023 and 2024. This chapter alone can be worth four marks.

Ranking – three formulas and the plus-one

All of ranking rests on one idea: when you count a person from both ends, that person gets counted twice. Every plus one and minus one in the formulas below is just correcting for that double count. Learn the three formulas as they stand and stop deriving them in the hall.

- Rank from the bottom = Total – rank from the top + 1.
- When both ranks belong to the SAME person: Total = rank from one end + rank from the other end – 1.
- When the two ranks belong to DIFFERENT persons who do not overlap: number between them = Total – rank of the first from the left – rank of the second from the right.
- Moving k places to the right: new position from the left = old position + k.
- After two persons interchange, each occupies the other's ORIGINAL seat – convert the new rank back to the old person's position first.

Ranking – what to write down

Read whether the two given ranks belong to one person or to two different persons.

→ If one person: add the two ranks and subtract 1 to get the total.

→ If two persons: convert both ranks to the SAME end – usually both from the left.

→ For a shift, apply the shift to the position from the left, then convert back.

→ For an interchange, remember the mover has taken over somebody else's old seat.

→ Sanity-check: no position may exceed the total, and none may be less than 1.

Worked example 1. In a row of children, Nita is 7th from the left end and Radha is 9th from the right end. When they interchange their places, Nita becomes 11th from the left end. How many children are there in the row?

- After the exchange, Nita is sitting in Radha's ORIGINAL seat. That is the key sentence.
- That seat is 11th from the left, because that is where Nita now is.
- The same seat was Radha's, and Radha was 9th from the right.
- So one single seat is 11th from the left and 9th from the right.
- Both ranks now belong to the same seat, so Total = 11 + 9 - 1 = 19.
- Answer: 19 children. Note that Nita's original 7th from the left was never needed - it is a deliberate distractor.

TRAP

Do not use Total = a + b - 1 when the two ranks belong to two different people. That formula is only for one person counted from both ends. Different people need the subtraction formula, and if they overlap in the row the answer changes again. Read the names before you pick the formula.

Comparison and ordering puzzles

Here you get four or five separate comparisons - A is taller than B, D is taller than C - and must find who is tallest or who is third. The mistake is to keep the pairs as pairs. Merge them into one single chain and the answer is just a matter of counting along it.

- Write every clue in the same direction. Convert 'X is shorter than Y' into 'Y > X'.
- Link the pairs into one chain: A > B > C and so on. Do not keep separate fragments.

- The tallest or heaviest sits at one end of the chain, the shortest or lightest at the other.
- Count from the end the question names - 'third heaviest' counts from the heavy end.
- If two people cannot be linked by any clue, the answer is 'cannot be determined'.

Worked example 2. Five boxes P, Q, R, S and T have different weights. P is heavier than S, R is heavier than P, Q is heavier than R, and T is lighter than S. Which box is the third heaviest?

- P is heavier than S gives P > S.
- R is heavier than P gives R > P, so the chain grows to R > P > S.
- Q is heavier than R gives Q > R, so Q > R > P > S.
- T is lighter than S gives S > T, so the full chain is Q > R > P > S > T.
- Count from the heaviest end: Q is 1st, R is 2nd, P is 3rd.
- Answer: P is the third heaviest.

Linear seating - fix a reference point before anything else

This is the heart of the chapter and it has a fixed procedure. Draw as many blank dashes as there are seats. Then hunt for the clue with the fewest possibilities - a clue that names an END seat or the exact MIDDLE seat. That clue is your reference point. Every other clue gets attached to it. If you start from a floating clue like 'D sits third to the right of A' you will be carrying four cases instead of one.

Seating arrangement – the fixed procedure

Count the seats and draw exactly that many dashes in a line. → Write down which way the persons face; if they face South, note the flip at once. → Scan every clue and pick the one that fixes an END or the MIDDLE seat. Place it. → Take the next strongest clue – the one that ties two named persons at a fixed gap. → Slide that block of two along the dashes and keep only the positions that fit. → Add the remaining positive clues one at a time; negative clues ('not next to') go LAST. → When one arrangement survives, re-read every clue against it before answering. → If two arrangements survive, you have misread a clue – go back, do not guess.

- For a row facing NORTH, the persons' right is towards the right of the page as you read it.
- For a row facing SOUTH, everything reverses: the persons' right is towards the LEFT of the page.
- 'X sits third to the left of Y' means exactly TWO persons sit between X and Y.
- 'n persons sit between X and Y' means the gap between their seats is n + 1.
- 'Immediately to the right of' means the very next seat on that person's right side.
- Write the row once in printed order and flip mentally; do not redraw it upside down.

Worked example 3. Eight students A, B, C, D, E, F, G and H sit in a straight row facing North. C sits third to the right of H. G is not an immediate neighbour of E. H is not an immediate neighbour of F. Only one person sits between G and H. D sits at the extreme left end. C is not an immediate neighbour of G. Two persons sit between B and F. G sits third to the left of A. Who sits exactly between H and F, and which pair holds the two ends?

- Start with the end clue: D is at the extreme left. Seat 1 = D. That is the reference point.
- Next take the linked pair: only one person sits between G and H, so G and H are two seats apart.
- G sits third to the left of A, so A sits three seats to the right of G. Try G at seat 2: then A is at seat 5.
- With G at seat 2, the gap clue puts H at seat 4 (H at seat 0 is impossible).
- C sits third to the right of H, so C is at seat 7.
- Two persons sit between B and F, a gap of three seats. The free seats are 3, 6 and 8. B at 3 and F at 6 fits.
- Seat 8 is left for E. Row from the left: D, G, B, H, A, F, C, E.
- Check the negatives: G at 2 is not beside E at 8; H at 4 is not beside F at 6; C at 7 is not beside G. All hold.
- H is at seat 4 and F at seat 6, so the person exactly between them is at seat 5 - A.
- The two ends are seat 1 and seat 8 - D and E.

TRAP

When a row faces South, the phrase 'immediately to the right of P' means the seat to the LEFT of P on your page. Students solve the whole arrangement correctly and then read the final answer off in printed order. Write a small arrow above the row showing which way the persons face, and read the answer off that arrow, not off the page.

Circular, square and rectangular tables

A circle has no fixed left end, so you must create one. Pick any person, draw him at the top of your circle, and place everybody else relative to him. Then the only thing that can go wrong is the direction of left and right - and that depends entirely on whether the group is facing the centre or facing outward.

Facing the centre versus facing outward - learn this pair together

ALL FACING THE CENTRE	ALL FACING OUTWARD
The clockwise neighbour is on a person's LEFT	The clockwise neighbour is on a person's RIGHT
The anti-clockwise neighbour is on the RIGHT	The anti-clockwise neighbour is on the LEFT
'Second to the left' = two seats clockwise	'Second to the left' = two seats anti-clockwise
'Third to the right' = three seats anti-clockwise	'Third to the right' = three seats clockwise
This is the default - assume it unless the stem says otherwise	Applies only when the stem explicitly says facing outward
The seat exactly opposite is n/2 seats away	The seat exactly opposite is still n/2 seats away

- Fix one person as the reference; a circle has relative order only, never absolute position.
- In a circle of n persons, the seat exactly opposite is n/2 seats away - defined only when n is even.
- Persons between X and Y counted clockwise = clockwise gap - 1; the other way it is n - clockwise gap - 1.
- The opposite-seat rule does NOT change with facing; only left and right change.

- At a SQUARE table with one person per corner, 'diagonally opposite' means two corners away, which is also second to the left and second to the right.
- At a rectangular table with two persons on each long side and one on each short side, treat the six seats as one ring and use the circle rules.

Worked example 4 (asked in RAS Pre 2021). Eight friends P, Q, R, S, T, U, V and W sit around a circular table facing the centre. S sits immediately to the right of P. W sits second to the right of Q. V sits third to the left of W. T sits fourth to the left of U. T sits third to the left of S. Who sits exactly opposite W, and who sits third to the right of Q?

- All face the centre, so translate first: 'to the right' means anti-clockwise, 'to the left' means clockwise.
- Fix S at the top of the circle as the reference point.
- S is immediately to the right of P, so S is one seat anti-clockwise from P; P sits immediately clockwise of S.
- T sits third to the left of S means T is three seats clockwise from S.
- T sits fourth to the left of U places U four seats clockwise from T, which is seven clockwise from S.
- Applying the Q and W clues the same way, exactly one seating survives.
- Clockwise order: S, P, W, T, Q, V, R, U, and back to S.
- Opposite W: with eight persons the opposite seat is four away. Counting four from W gives R.
- Third to the right of Q: right means anti-clockwise. Step anti-clockwise from Q - first T, second W, third P.
- Answers: R sits opposite W; P sits third to the right of Q.

Double rows, floors, days and months

A double-row puzzle has five persons in row 1 facing North and five in row 2 facing South, sitting directly opposite each other. Because the two rows face each other, their left-right senses are mirror images. Write both rows in printed left-to-right order, one above the other, and put a small arrow on each row showing the facing. Then never redraw.

- Row 1 faces North: their right is the printed right. Row 2 faces South: their right is the printed LEFT.
- The person at printed position k in row 1 faces the person at printed position k in row 2.
- If X faces Y, then the person on X's right faces the person on Y's LEFT - this is the double-row rule.
- Solve the row that has the strongest clue first, then transfer through the 'X faces Y' clue.
- Always name the final answer using the facing of the row the question asks about.

Worked example 5. Ten persons sit in two parallel rows of five. A, B, C, D and E sit in row 1 facing North; P, Q, R, S and T sit in row 2 facing South; each row-1 member faces exactly one row-2 member. A sits immediately to the right of C. Three persons sit between C and D. Only one person sits between A and B. Only one person sits between T and R. C faces S. R sits third to the left of Q. Who faces B, and who sits at the extreme right end of the second row?

- Row 1 faces North, so for row 1 the printed right is their right. Three persons between C and D means C and D are four seats apart, i.e. seats 1 and 5.
- A sits immediately to the right of C. If C is at printed seat 1, A is at printed seat 2.
- Only one person sits between A and B, a gap of two seats, so B is at printed seat 4. E takes seat 3.
- Row 1 printed left to right: C, A, E, B, D.

- Row 2 faces South, so for row 2 their left is the printed right. R sits third to the left of Q places R three printed seats to the RIGHT of Q.
- C faces S, and C is at printed seat 1, so S is at printed seat 1 of row 2.
- R = Q + 3 in printed terms, and seat 1 is taken by S, so Q must be at printed 2 and R at printed 5.
- Only one person sits between T and R, a gap of two seats, so T is at printed 3. P takes the last free seat, printed 4.
- Row 2 printed left to right: S, Q, T, P, R.
- B is at printed seat 4 of row 1, so the person facing him is at printed seat 4 of row 2 - P.
- Extreme right end of the SECOND row: that row faces South, so their right is the printed left - seat 1, which is S.

- FLOOR puzzles: always draw floor 1 at the bottom and the top floor at the top. 'Above' means a larger floor number.
- 'Only k persons live between X and Y' fixes the gap but not who is higher - test both cases before rejecting either.
- DAY and MONTH puzzles: number the days or months as slots 1 to n, then convert every 'before' and 'after' clue into a difference of slot numbers.
- 'R visits two days after P' means slot of R = slot of P + 2, not 'somewhere later'.
- Verify the finished arrangement by reading every clue once more; if a second valid arrangement exists, you have mis-read a clue.

YAAD RAKHO

Yaad rakho for circles - CLOCK LEFT. When everyone faces the CentrE, CLockwise is LEft. Flip both words for facing outward. Two words, and the biggest circular-seating error is closed.

Logical Venn diagrams - picking the right picture

This form gives you three class names - say Doctors, Men and Human beings - and four candidate pictures. You choose the one that shows the real relation. There are only three drawing rules, and only about five pictures RPSC ever uses, so this is close to free marks once you have seen the standard patterns.

- If EVERY A is a B, draw the circle A wholly inside the circle B.
- If SOME A are B, draw two circles that partly overlap.
- If NO A is a B, draw two completely separate circles.
- Ask the two questions in order: can something be both? and must everything of one be the other?
- Test each option against a real example before choosing - one counter-example kills a picture.

The five standard patterns RPSC reuses

Pattern	Example	Picture
Fully nested	Jaipur, Rajasthan, India	Smallest inside middle inside largest
Co-ordinate inside a whole	Table, Chair, Furniture	Two separate circles both inside a bigger one
One nested, one separate	Pigeon, Bird, Dog	Pigeon inside Bird; Dog a separate circle
Two overlapping inside a whole	Doctors, Men, Human beings	Doctors and Men overlap, both inside Human beings
Two separate, both overlapping a third	Boys, Girls, Students	Boys and Girls disjoint, Students cuts across each

ASKED IN RAS

Asked in RAS Pre 2023 - the relation between Doctors, Men and Human beings. Answer: Doctors and Men are two intersecting circles, both lying wholly inside a larger circle for Human beings, because some doctors are women. The same form came in 2024 (Jaipur, Rajasthan, India - fully nested), in 2021 (Boys, Girls, Students) and in 2016. Memorise the five patterns above and this question is worth a mark in ten seconds.

Numeric Venn - the two-set and three-set formulas

The numeric form gives you how many people do each of two or three things and asks how many do both, or none, or exactly one. It is pure inclusion-exclusion. You add the singles, then take out what you double-counted, then put back what you removed twice. Write the formula on the sheet before you substitute a single number.

- TWO SETS: n(A or B) = n(A) + n(B) - n(A and B).
- Those in NEITHER = Total - n(A or B).
- THREE SETS: n(A or B or C) = n(A) + n(B) + n(C) - n(A and B) - n(B and C) - n(A and C) + n(A and B and C).
- Exactly ONE of three = sum of the singles - 2 times the sum of the pairwise + 3 times all three.
- Exactly TWO of three = sum of the pairwise - 3 times all three.
- If the paper says everybody does at least one thing, then n(A or B) equals the total - use that to find the overlap.

Worked example 6 (asked in RAS Pre 2016). In a class of 60 students, 35 play cricket, 25 play hockey and 10 play both games. How many students play neither game?

- Write the two-set formula first: n(A or B) = n(A) + n(B) - n(A and B).
- Substitute: 35 + 25 - 10 = 50 students play at least one game.
- The 10 who play both were counted once in the 35 and once in the 25, so removing 10 once is exactly right.
- Neither = Total - at least one = 60 - 50 = 10.
- Answer: 10 students play neither game.

Worked example 7 (asked in RAS Pre 2024). Out of 200 students, 100 study Mathematics, 90 Physics and 80 Chemistry. 40 study both Mathematics and Physics, 35 both Physics and Chemistry, 30 both Mathematics and Chemistry, and 20 study all three. How many study none of the three?

- Write the three-set formula before substituting anything.
- Sum of singles: 100 + 90 + 80 = 270.
- Sum of pairwise: 40 + 35 + 30 = 105. Subtract it: 270 - 105 = 165.
- Add back the all-three figure once: 165 + 20 = 185 study at least one subject.
- None = 200 - 185 = 15.
- Answer: 15 students. If the question had asked for exactly one instead: 270 - 2(105) + 3(20) = 270 - 210 + 60 = 120.

TRAP

In set questions the pairwise figures normally INCLUDE the people who do all three. '40 study both Mathematics and Physics' means 40 in total, of whom 20 also do Chemistry. Only if the paper actually prints the word 'only' does the figure exclude them. Read for that one word before you substitute - the whole answer turns on it.

REVISION RECAP

1. Rank from the bottom = Total - rank from the top + 1; the +1 corrects the double count.

2. Both ranks of the SAME person: Total = rank from one end + rank from the other end - 1.
3. Different persons: convert both ranks to the same end before subtracting; comparison puzzles merge into one inequality chain.
4. After an interchange, the mover occupies the other person's original seat - use that seat's two ranks.
5. Seating: draw the dashes, then start from the clue that fixes an END or the MIDDLE seat.
6. Negative clues such as 'not an immediate neighbour' are used last, to eliminate, never to start.
7. 'Third to the left of Y' means exactly two persons between; 'n persons between' means a gap of n+1.
8. Row facing South: the persons' right is the printed left. Mark the facing with an arrow.
9. Double row: row 1 North and row 2 South face each other, so their left-right senses are mirrored.
10. In a double row, if X faces Y then the person on X's right faces the person on Y's left.
11. Circle facing the CENTRE: clockwise is LEFT, anti-clockwise is RIGHT. Facing outward reverses both.
12. In a circle of n persons the opposite seat is n/2 away, whichever way they face; at a square table diagonally opposite is two corners.
13. Floors: number 1 at the bottom; a gap clue does not say which of the two is higher, so test both.
14. Day and month puzzles: number the slots 1 to n and turn every before/after clue into a slot difference.
15. Logical Venn: all A are B means A inside B; some means overlap; no A is B means separate circles.
16. Two sets: n(A or B) = n(A) + n(B) - n(A and B); neither = Total - n(A or B).
17. Three sets: add the singles, subtract all three pairwise, add back the all-three figure.
18. Pairwise figures include the all-three people unless the paper prints the word 'only'.